Manuel Oelmaier

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Werkstudent at Agile Robots SE, Munich (20 h/week from Sep. 2026; M.Sc. lectures from 15 Oct)

Profile

B.Sc. Computer Science (TUM), thesis submitted Sept. 2026; M.Sc. Robotics, Cognition, Intelligence — lectures from Oct. 2026 (AI/ML focus). Currently Werkstudent at Agile Robots SE (Munich, 20 h/week). Bachelor thesis on token eviction in linear-attention transformers. Production infra (FIDE Candidates 2026), code-gen finetuning with eval harness, self-hosted AI platform with RAG and MCP agents. Python, Linux/HPC, TypeScript.

Education

Master of Science – Robotics, Cognition, Intelligence from Oct. 2026
Technical University of Munich, Germany

- Focus: AI, machine learning, intelligent systems

Bachelor of Science in Computer Science 2021 – Sept. 2026
Technical University of Munich, Germany

- Studies ongoing – expected graduation September 2026
- Focus: Machine Learning, Artificial Intelligence, Software Development
- Relevant coursework: Praktikum Next Gen AI, Ethics in Artificial Intelligence, Fundamentals of Programming
- *Thesis (submitted):* **Hidden State Informed Token Eviction**

Abitur (secondary school diploma) 2013 – 2021
Oskar-Maria-Graf Gymnasium, Germany

- *Seminar paper:* Minimax-based AI for a self-created game
  Before the current AI hype

Professional Experience

Working Student – Agile Robots SE Sep. 2026 – present
Munich. Werkstudent, 20 h/week. AI models for robotics (VLMs, world models), workcell semantic layer, and agentic workflows connecting language/multimodal models with robotics tools.

Infrastructure Engineer – Team Sindarov / FIDE Candidates 2026 2025 – 2026
One of two engineers behind the distributed engine-analysis infrastructure used throughout preparation for and during the FIDE Candidates Tournament 2026. Owned setup, deployment, and day-to-day technical communication with players and seconds.

- Co-owned hardware selection and performance tuning to maximize sustained engine throughput
- Hardened security: secure, access-controlled remote analysis for confidential opening preparation
- Built an internal observability and control dashboard: live status, cost-aware on-demand scaling, audit logs
- Primary technical point of contact: onboarding, training, troubleshooting, feature requests

- **Impact:** sustained high-throughput analysis with cost-aware on-demand scaling; zero unplanned downtime; team won the Candidates with a record 10/14

Projects

Bachelor thesis (submitted): Hidden State Informed Token Eviction

- Efficient LLM inference: learnable KV-cache token eviction routed by a linear-attention memory's reconstruction-error signal
- Benchmark harness with accuracy/throughput ablations; hands-on PyTorch implementation and evaluation

Praktikum Next Gen AI: LLM finetuning & VS Code extension 2025

- Finetuned transformer models for Python code generation; built an automated multi-case evaluation pipeline
- Improved model performance from 40 % to 75 % on code-generation tasks
- Published the best model as a VS Code extension for local, privacy-preserving in-editor queries

Production AI platform – Homelab 2025 – ongoing

- Self-hosted stack (Docker, Ollama, LibreChat, MCP agents); **RAG pipeline** over local documents, context injection into agent/chat flows
- Basic outline on my site: manueloelmaier.de/blog/homelab-ai-platform

Learning platform – Theoretical Informatics (IN0011) 2025

- Aggregated past TUM exam questions into an interactive quiz platform
- Accurate rendering of mathematical formulas and notation

ManuChess – playing-style mirror via neural network 2025 – ongoing

- Pipeline from Lichess games to (position, move) training pairs in a database
- Training a neural network to predict my next move from a board position

Skills

Technical: Python, Java, JavaScript/TypeScript, HTML, CSS, C, Bash, SQL, MongoDB
AI / ML: PyTorch, Hugging Face, LLM finetuning & evaluation, sparse attention, KV-cache eviction, LLM inference efficiency (thesis), MCP
Systems / Infra: Linux, Git, Docker, distributed/HPC workloads, performance tuning (CPU affinity, NUMA), SSH & network security, observability dashboards, cost-aware cloud scaling
Languages: German (native), English (full professional proficiency, C1; English-taught M.Sc.)
Soft skills: problem-solving, communication, discretion with sensitive data, end-to-end ownership

Volunteering & Hobbies

- Volunteer youth chess trainer, SC Eching
- Competitive tournament chess player

References

IM Roman Vidonyak – Chief Trainer, Team Sindarov (winner of the FIDE Candidates 2026 and challenger for the World Chess Championship); Head Trainer, FC Bayern Munich Chess
Introduction and recommendation letter on request.